

Background

In past attempts people have used foveated rendering (distance from gaze) in VR to reduce GPU. However, it's not optimal for human experience. This method uses perceptual gaussian splatting (content aware), a new method to optimize rendering that is energy efficient and improves the user experience.

Methods

Instead of using foveated rendering where clarity is reduced based on distance from gaze, perceptual gaussian splatting was chosen as the power saving method, which focuses on objects in scene content.

3D gaussian splatting (3DGS) renders the scene using gaussians (blobs) instead of pixels. It also does rasterization, or blurring edges, which is more cost efficient and effective compared to ray tracing, which has the computer trace every single pixel.

Test scenes were created using Blender. COLMAP uses novel-view synthesis to reconstruct 3D scenes through neural networks (Figure 1).

The hierarchical structure and the eye tracker allows the clarity of the scene to change in real time to optimize energy while ensuring user experience. (Figure 2, 3, 4)

References

1. Bernhard Kerbl, Andreas Meuleman, Georgios Kopanas, Michael Wimmer, Alexandre Lanvin, and George Drettakis, "A Hierarchical 3D Gaussian Representation for Real-Time Rendering of Very Large Datasets," <https://repo-sam.inria.fr/fungraph/hierarchical-3d-gaussians/>
2. Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, George Drettakis, "3D Gaussian Splatting for Real-Time Radiance Field Rendering," <https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/>

Content-Aware Rendering for Efficient Virtual Reality

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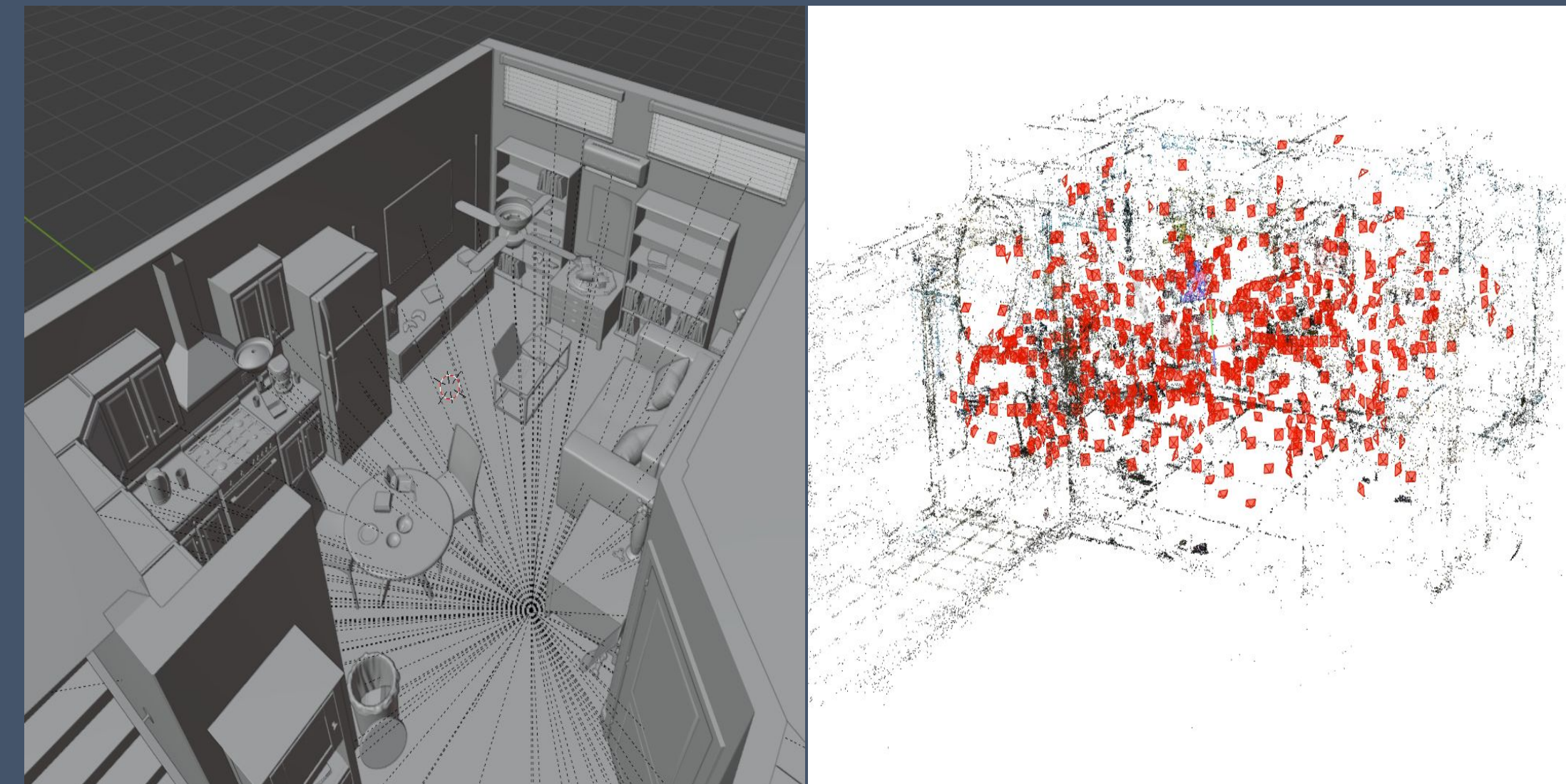


Figure 1. Image of Blender scene and COLMAP 3D scene reconstruction. Red dots are cameras, black outline is computer generated.



Figure 3. Image of the scene fully rendered. Best outcome

Pros: Clear visualization of the scene

Cons: This is a display of the 3D Model on the computer, so implementing it on VR would be extremely costly



a) Scene rendered with peripheral vision blurred - saving GPU
 b) Scene rendered with everything in focus - requiring more GPU
 Figure 2. A comparison of the same image with different levels of rendering. Fast and efficient for real time focus change.



Figure 4. Rendering of the scene in VR using hierarchical 3DGS

Pros: Cost efficient and optimizes GPU

Cons: Gaussians on the edges can be blurry, and lighting can be weird



Results

After creating a testing interface in Unity, participants were given a reference video and shown two options using different rendering techniques. They decided which option was most similar to the reference video.

Participants reported a better experience with the perceptual gaussian splatting method than traditional methods using foveated rendering or uniform rendering (Figure 5).

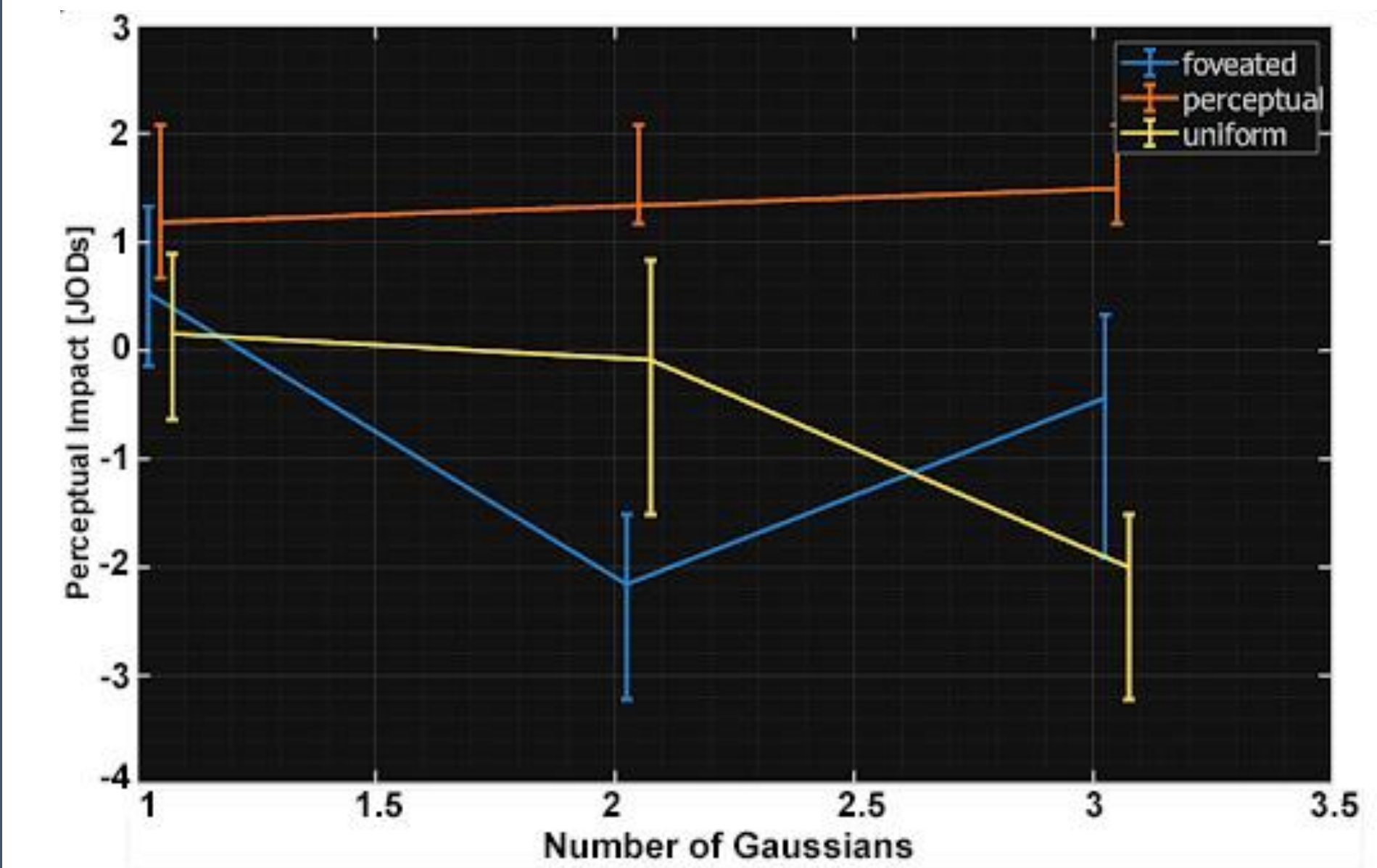


Figure 5. Graph comparing the different rendering methods (foveated, perceptual, and uniform) in Number of Gaussians and Perceptual Impact in Just-Objectable-Differences (JODs), user preference. Perceptual rendering was most preferred.

Future Works

Developing a gaze-agnostic approach, since eye tracking is computationally expensive, is a potential area for future work. The AI program would intelligently decide where to allocate detail without needing to know exactly where the user is looking.

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